Define Problem Statements

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

1. Problem Context

In lecture halls with 60 to 120 students, taking attendance by roll call costs 5 to 10 minutes per lecture, and paper sheets or card taps can be faked by friends. At the same time, lecturers have no objective feedback on whether students are paying attention. Existing biometric solutions often send video of students to the cloud, raising privacy and legal concerns (GDPR, FERPA, and India's Digital Personal Data Protection Act 2023).

2. Problem Statement

How might we automatically record verified classroom attendance and measure student engagement in real time, using only a standard laptop and camera, **without** uploading any video or biometric data to the cloud and **with** explicit student consent?

3. Problem Statement Canvas

I am	Trying to	But	Because	Which makes me feel
A university lecturer teaching a large class	Record attendance quickly and know whether students are following the lecture	Roll calls waste time, proxies slip through, and I cannot see who has drifted off	Manual methods and card systems do not verify presence over time, and I have no attention data	Frustrated, uncertain, and short of teaching time
A student in the class	Have my attendance recorded fairly and my privacy respected	I worry about cameras uploading my face somewhere unknown	Many systems are cloud-based and do not ask for clear consent	Uneasy and watched
A department administrator	Keep reliable attendance records and exports	Proxy attendance corrupts the data, and edits are not audited	Paper and tap systems have no audit trail	Concerned about accuracy and compliance

4. Detailed Problem Statements

ID	Problem	Who is affected	Evidence / consequence
PS1	Manual attendance is slow.	Lecturers, students	5 to 10 minutes lost per lecture in classes of 60-120.
PS2	Proxy attendance and buddy punching.	Institution, honest students	Paper sheets and card taps can be done on someone else's behalf.
PS3	No objective measure of attention.	Lecturers	Cannot tell when students are overloaded, distracted, or disengaged.
PS4	Cloud surveillance risk.	Students	Uploading video or biometrics to remote servers conflicts with privacy laws and trust.

5. Success Criteria

- Attendance computed from verified presence time, not a one-time snapshot.
- No raw video or biometric template leaves the local machine; unconsented people are never enrolled.
- A single engagement index per student and for the class, updated in real time.
- Actionable teaching suggestions without alert spam (30-second debounce).
- Lecturer can correct any record with a reason, and export a CSV of results.

6. Scope Boundaries

In scope: single camera per session, face detection/tracking/recognition, head pose and gaze, audio classification, fusion, attendance, recommendations, local database, dashboard. **Out of scope:** measuring cognitive understanding or emotion, multi-camera auditorium coverage, and cloud services.